NEO CHALLENGE: DAILY CHALLENGE

One app, three daily habits: train your body, grow your languages, sharpen your mind.

Software project documentation · AFFIAT · Afro-Asian Forum for Innovation and Technology · Aswan Finals 2026 · Software Category

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1. Project Overview | نظرة عامة على المشروع

Field	Value
Project Title	NEO Challenge
Project Type	Web App (Python + Streamlit)
Description	NEO Challenge is a free web app that puts three small daily habits in one place: sports challenges, language activities (English, French, German) and a general knowledge quiz. A student picks a challenge, joins it, does a short activity, and comes back tomorrow. It also has a workout and exercise library with 60 demonstration videos, a muscle explorer, and nutrition plans by BMI category.
Target Users	Primary: students aged 12-18. Secondary: teachers, parents and schools. Tertiary: sports clubs and youth / language centres.

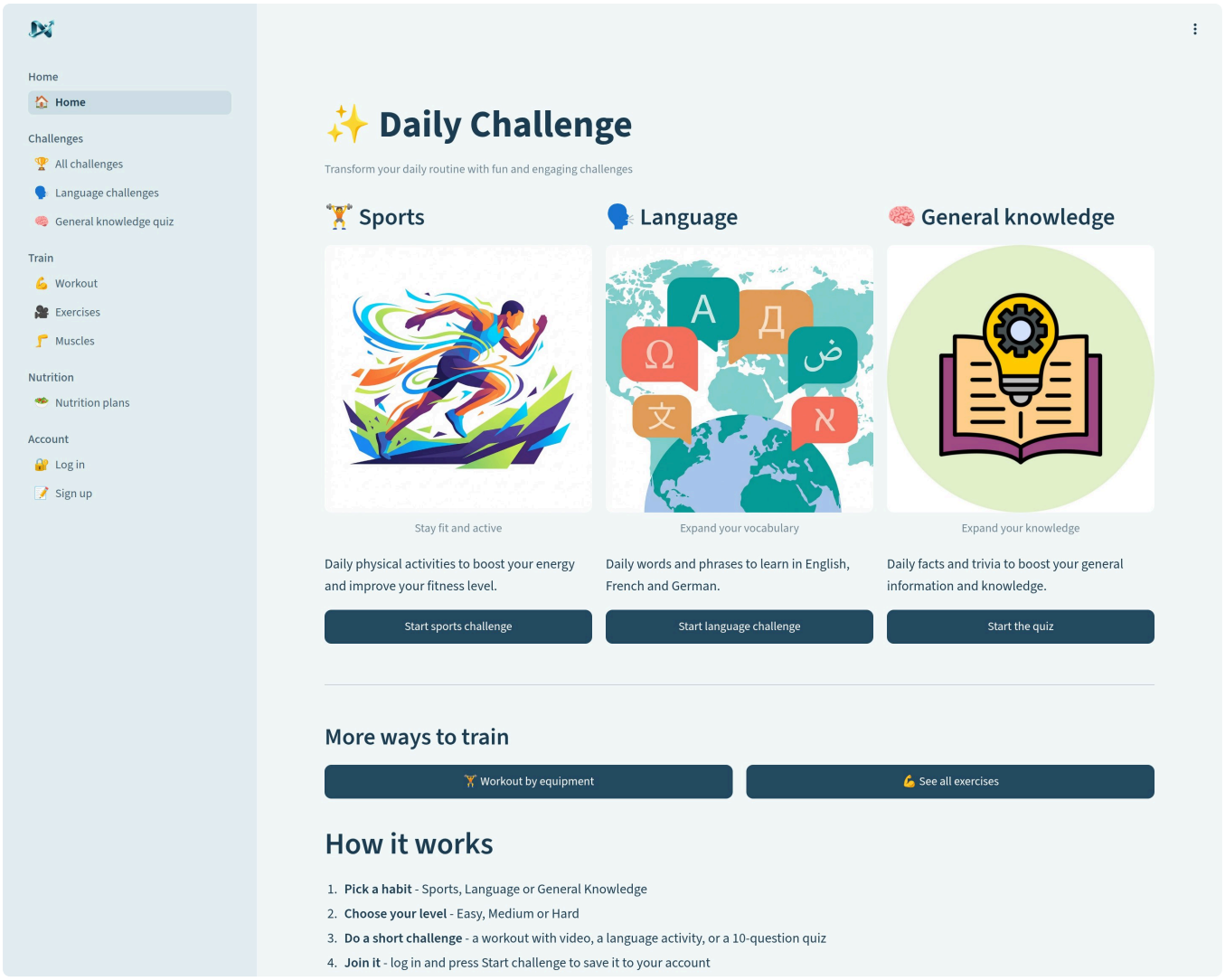
Field	Value
Resolution / Platform	Runs in any modern browser. Desktop-first wide layout, tested at 1440x900. Streamlit columns stack on narrow widths, so mobile is partially tested.
Date	October 2026
Live app	https://neo-challenge.streamlit.app/

Tech Stack | التقنيات المستخدمة

Category	Technology	Purpose
Language	Python 3	All application logic: screens, database queries, the menu
Framework	Streamlit 1.64	Builds every screen, the menu (`st.navigation`) and remembers who is logged in (`st.session_state`)
Database	SQLite via Python's built-in `sqlite3`	Stores users, challenges, activities, questions, muscles, exercises and nutrition plans in `database/database.db`
Libraries	Streamlit only	`requirements.txt` contains exactly one package, `streamlit==1.64.0`. `sqlite3` and `datetime` come with Python.
Styling	Streamlit theme (`.streamlit/config.toml`)	One teal colour palette for the whole app, no custom CSS

Dependencies | المكتبات المستخدمة

Package	Version	Purpose
streamlit	1.64.0	The whole user interface: widgets, menu, session state, video and audio playback
gTTS	2.5.4	Not part of the app. `tools/make_audio.py` used it once to make the 90 audio clips; the app only plays the saved mp3 files.



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2. Problem Statement | بيان المشكلة

What problem does this project solve? | ما المشكلة التي يحلها هذا المشروع؟

Teenagers who want to improve themselves usually spread their effort across several separate apps: a generic fitness tracker for the gym, a language app for vocabulary, and a quiz app for trivia. Each one is built for adults, most are full of ads or paywalls, and none of them help a young person keep a *small* daily habit going. NEO Challenge gives a student one simple, free place to build three small habits a day: train the body, grow the languages, and sharpen the mind.

Why does it matter? | لماذا هذا مهم؟

Students aged 12-18 are exactly the age group that most needs daily movement, language practice and mental stimulation, and the age group for whom a habit is easiest to build when it is short and in one place. Without one place to return to, effort never becomes a habit. Teachers and parents have the same problem from the other side: there is no free tool they can recommend without an adult account.

How is it currently solved? | كيف تُحل المشكلة حاليًا؟

Students use separate apps (generic fitness trackers, language-learning apps, trivia apps) or simply follow YouTube workout videos with no structure. The separate apps are designed for adults, are ad-heavy or overwhelming, and cost money. The YouTube route has no plan, no levels and no way to remember what you joined. Neither route puts the three habits together.

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3. Technical Architecture | البنية التقنية

Project Structure | هيكل المشروع

```
neo_challenge_database/
├── app.py                       # Creates the database, builds the menu (82 lines)
├── requirements.txt             # One dependency: streamlit==1.64.0
├── README.md                    # How it works, how to run it, Limits
├── .streamlit/config.toml       # The app's colours
├── assets/
│   ├── config.py                # Database path, pictures, level icons, voice languages
│   ├── thumbnails/              # A picture from each exercise video
│   └── *.jpg / *.png / *.webp   # Category, flag, muscle and equipment pictures
├── app_pages/                   # 20 screens, one plain script each
│   ├── home.py                  # Home: the three habits
│   ├── view_challenges.py       # All challenges
│   ├── language_challenge.py    # English / French / German challenges
│   ├── challenge_details.py     # One challenge: join it, activities by level
│   ├── general_knowledge.py     # Quiz: level, 10 random questions, score
│   ├── workout.py               # Pick your equipment
│   ├── view_exercise.py / exercise_details.py   # Fitness level, exercises + video
│   ├── view_muscles.py          # Muscles with anatomy facts
│   ├── view_nutrition_plan.py   # BMI weekly plan, nutrition plans, choose your plan
│   ├── my_challenges.py         # What you joined and your plan
│   ├── login.py / sign_up.py / logout.py
│   ├── view_users.py            # Admin: everyone who signed up
│   └── create_*.py              # Admin: add challenges, activities, exercises,
│                                #        muscles and nutrition plans
├── database/
│   ├── schema.sql               # The 12 tables, with their rules
│   └── insert_*.sql             # Our starting content
├── videos/                      # 60 exercise demonstration clips (.mp4)
├── audio/                       # 90 clips that read the language sentences aloud (.mp3)
├── tools/
│   └── make_audio.py            # Makes the audio clips once with gTTS (not part of the
app)
```

Architecture Diagram | مخطط البنية

HOW THE MENU IS BUILT

EVERY RERUN

app.py runs first

Creates the tables if they are missing, fills an empty database, then builds the menu.

SESSION

Who is here?

"user_id" in st.session_state?
st.session_state.is_admin?

ST.NAVIGATION

Only allowed pages go in

A page that is not in the menu cannot be opened at all.

> The first person to sign up becomes the admin. The Challenge and Exercise pages are always there but hidden: buttons open them.

WHO SEES WHICH PAGE

Page	Guest not logged in	Student logged in	Admin is_admin = 1
EVERYONE			
🏠 Home	✓	✓	✓
🏆 All challenges · 🗣️ Language	✓	✓	✓
🧠 General knowledge quiz	✓	✓	✓
💪 Workout · 🎥 Exercises · 🦵 Muscles	✓	✓	✓
🥗 Nutrition plans	✓	✓	✓
GUESTS ONLY			
🔐 Log in · 📝 Sign up	✓	—	—
LOGGED IN			
⭐ My challenges	—	✓	✓
🚪 Log out	—	✓	✓
ADMIN			
👥 Users	—	—	✓
➕ Create challenge · activity	—	—	✓
➕ Create exercise · muscle	—	—	✓
➕ Create nutrition plan	—	—	✓

In plain words: one Python process serves every screen. `app.py` sets everything up when the app starts, so nobody has to run a separate setup script, and the menu only shows the screens you are allowed to open. Each screen is one short file that talks to the database with plain SQL and passes the selected challenge or exercise to the next screen through `st.session_state`.

Key Files | الملفات الرئيسية

File	Purpose	Lines of Code
app.py	Creates the tables, fills an empty database with our content, and builds the menu: Log in / Sign up when logged out, My challenges / Log out when logged in, Admin only for the admin.	82
app_pages/challenge_details.py	The busiest screen. Shows one challenge, counts its activities, lets a logged-in student join it, filters the activities by Easy / Medium / Hard, and plays the audio clip of each language sentence.	151
app_pages/general_knowledge.py	The quiz. Picks 10 random questions for the chosen level, shows them one at a time with right or wrong straight away, then the score and a ✅ / ❌ review, and saves the round for logged-in students.	132
app_pages/view_nutrition_plan.py	Works out your BMI from weight and height and shows a 7-day meal plan for your category, then lists the nutrition plans by BMI; a logged-in student can choose one plan.	120

File	Purpose	Lines of Code
app_pages/sign_up.py	Creates an account. Checks for empty fields and a used email; the very first person to sign up becomes the admin.	49
app_pages/view_exercise.py	A fitness level calculator (push-ups, squats and plank), then the exercise library, filtered by equipment, level and muscle.	93
app_pages/workout.py	Four equipment cards that open the filtered exercise library.	26
database/schema.sql	The 12 tables with their foreign keys and `CHECK` rules.	131

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4. Features & User Flow | المميزات ورحلة المستخدم

Feature List | قائمة المميزات

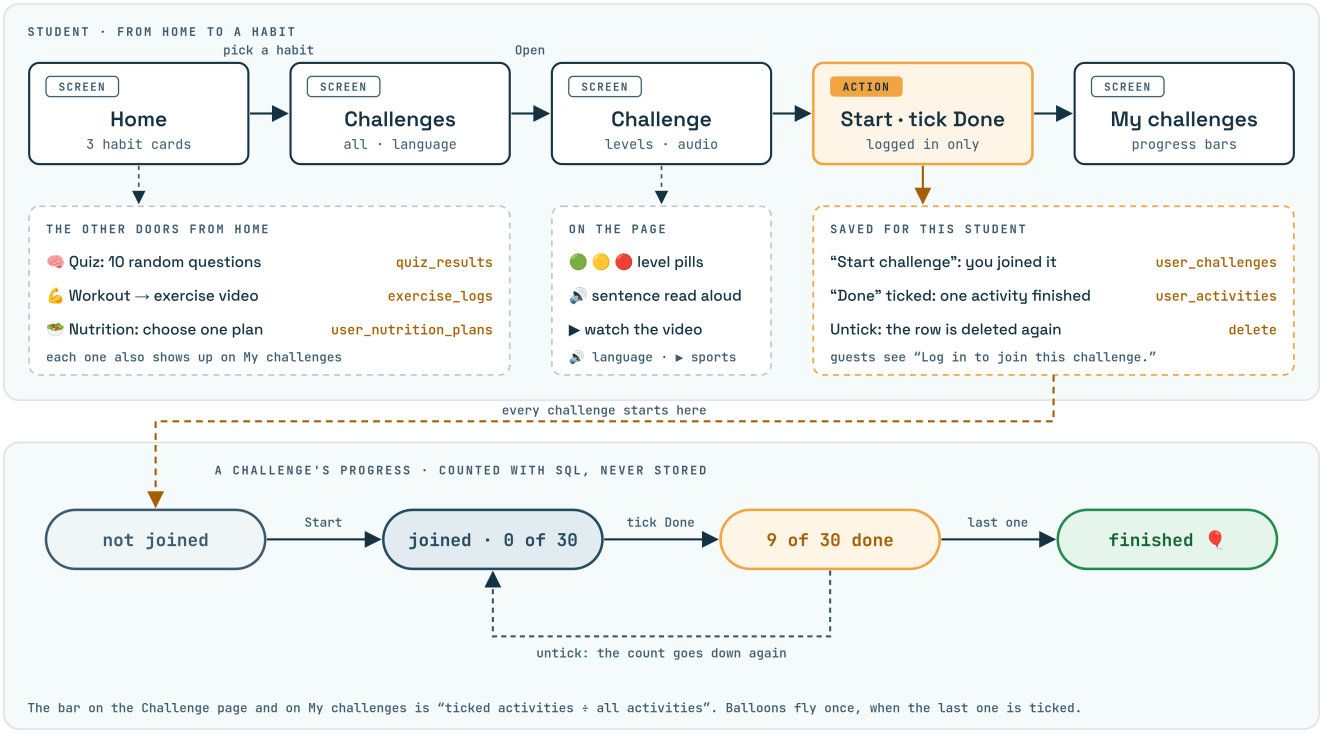
#	Feature	Description	Status
1	Home hub	One landing screen with three habit cards (Sports, Language, General knowledge) and shortcuts to Workout and Exercises.	✅
2	Challenge browser	All 12 challenges as cards with category, duration and a live count of their activities.	✅
3	Challenge page with level filter	Details, metrics and the challenge's activities, filtered by All / Easy / Medium / Hard.	✅
4	Join a challenge	A logged-in student presses "Start challenge" to save it; logged-out visitors see "Log in to join this challenge."	✅
5	Language challenges	English for Beginners, French Conversation and German For Beginner, each with 30 sentences (10 Easy, 10 Medium, 10 Hard) written by our team, and an audio clip that reads each sentence aloud.	✅
6	General knowledge quiz	Choose a level, then answer 10 random questions one at a time with a progress bar; each answer shows right or wrong straight away, and the end shows the score and a ✅ / ❌ review.	✅
7	Workout by equipment	Machines, Free weights, Bodyweight and Cables cards that open the matching exercises.	✅
8	Exercise library with videos	All 60 exercises have a picture taken from their video, a demonstration video, the muscle they train, the equipment and the difficulty; filter by equipment, level and muscle. Above the library, a fitness level calculator turns your push-ups, squats and plank time into Easy, Medium or Hard.	✅
9	Muscle explorer	15 muscles with pictures and an "Anatomy facts" section (Latin name, location, what it does, and more).	✅

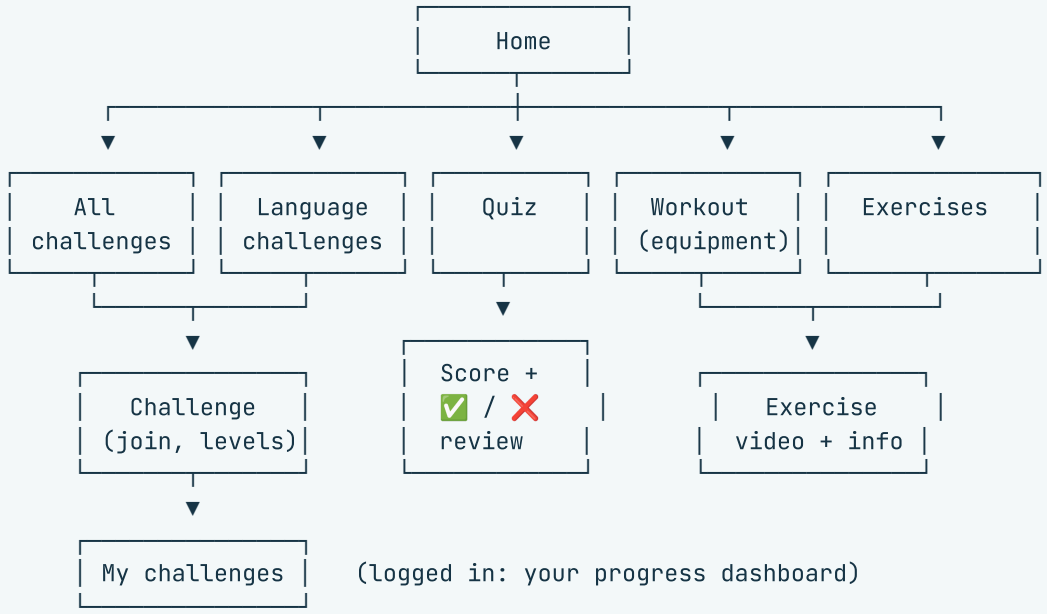
#	Feature	Description	Status
10	Nutrition plans	Type your weight and height to get your BMI and a 7-day meal plan (breakfast, lunch and dinner). Below it, 15 plans across Underweight, Normal, Overweight and Obese with calories, meals and water; a student can choose one.	✅
11	My challenges (progress)	A dashboard: a progress bar for every joined challenge, exercises done this week, best quiz score per level, recent rounds and the chosen nutrition plan.	✅
12	Sign up / log in / log out	Student accounts; the menu changes to show only what you can open.	✅
13	Admin pages	The first person to sign up is the admin and can see all users and add challenges, activities, exercises, muscles and nutrition plans.	✅
14	Password hashing	Passwords are stored in plain text today.	⏳
15	Progress tracking	Tick activities Done on a joined challenge, log an exercise once a day, and save every quiz round.	✅
16	Activities for Sports challenges	6 activities in each of the first 3 Sports challenges; the workout ones open the matching exercise video. The 5 newer Sports challenges have none yet.	⏳
17	Public deployment	Putting the app online at neo-challenge.streamlit.app, so any school can open it without installing Python.	⏳

> Status: ✅ Complete | ⏳ In Progress | ❌ Not Started

Known limitations (the full list is in `README.md` under Limits): passwords are stored in plain text; the app has no error-handling code beyond friendly checks for empty fields and used emails or names; progress has no reminders or day-in-a-row streaks yet; the language audio is made ahead of time with a computer voice; the fitness level and the 7-day meal plan are not saved, and the meal plan has three versions (a BMI of 30 or more gets the Overweight one); and the app is not deployed publicly.

User Flow | رحلة المستخدم





The sidebar menu is always there. It shows Log in / Sign up when you are logged out, My challenges / Log out when you are logged in, and the Admin pages only to the admin.

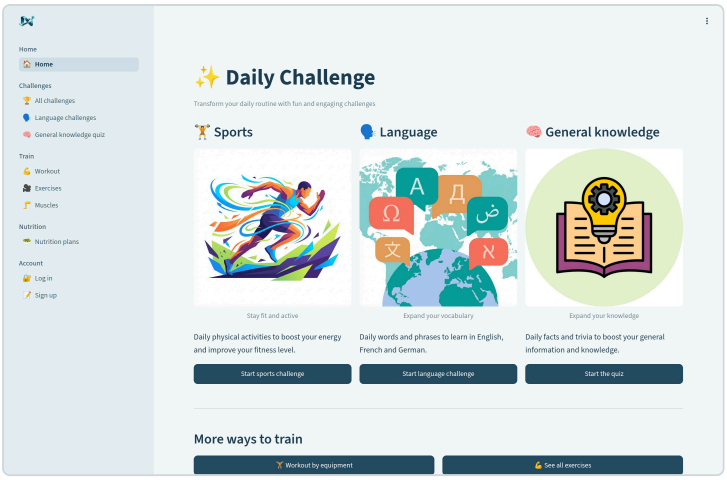
Screen Reference | الشاشات

Screen	File	Description	Transitions To
Home	app_pages/home.py	”Daily Challenge” headline, three habit cards, ”More ways to train”	All challenges, Language challenges, Quiz, Workout, Exercises
All challenges	app_pages/view_challenges.py	Every challenge with category, duration and activity count	Challenge
Language challenges	app_pages/language_challenge.py	English, French and German challenge cards with flags	Challenge
Challenge	app_pages/challenge_details.py	Picture, metrics, description, join button, activities by level, audio for language sentences	Quiz (General Knowledge only)
General knowledge quiz	app_pages/general_knowledge.py	Level pills, one question at a time, score and review	Play again on the same screen
Workout	app_pages/workout.py	Four equipment cards	Exercises (filtered)
Exercises	app_pages/view_exercise.py	Fitness level calculator, equipment and level pills, muscle filter, picture cards	Exercise
Exercise	app_pages/exercise_details.py	Muscle, equipment, difficulty, description and video	Exercises
Muscles	app_pages/view_muscles.py	Muscle pictures with ”Anatomy facts”	Workout
Nutrition plans	app_pages/view_nutrition_plan.py	BMI form with a 7-day meal plan, plan cards with meals; ”Choose this plan” when logged in	Stays on the page
My challenges	app_pages/my_challenges.py	Progress dashboard: challenges, workouts, quiz scores, nutrition plan	Challenge
Log in / Sign up / Log out	app_pages/login.py, sign_up.py, logout.py	Accounts	The menu changes
Users, Create … (admin)	app_pages/view_users.py, create_*.py	User list and forms to add content	—

5. UI/UX Design | تصميم واجهة المستخدم

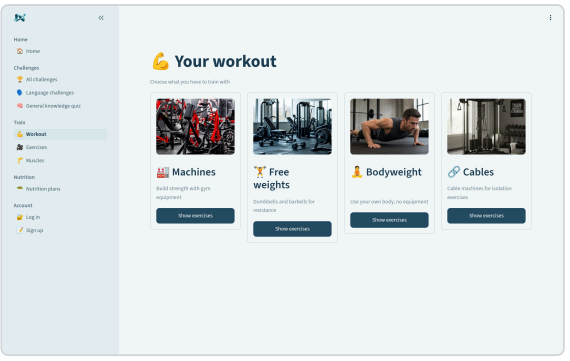
Screenshots | لقطات الشاشة

Home



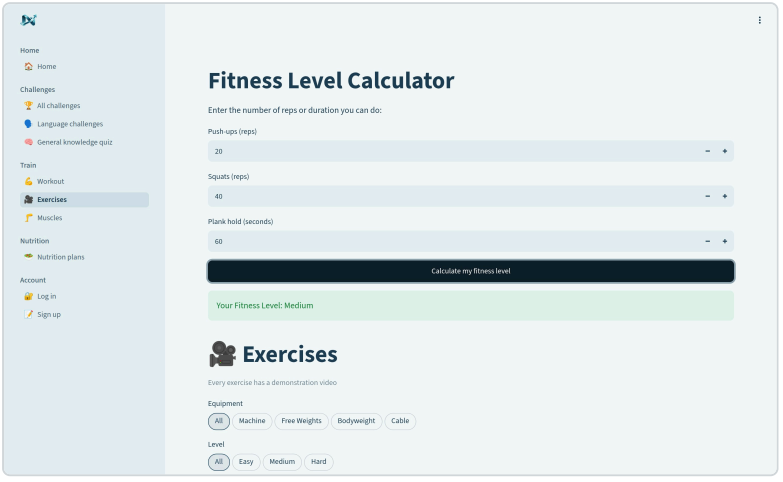
The three habits, each with one button, plus shortcuts to Workout and Exercises.

Workout



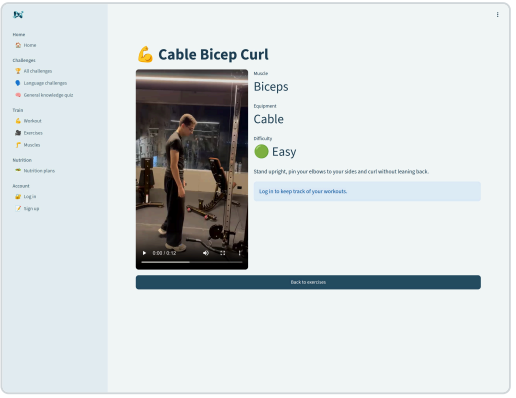
One card per equipment type.

Exercises



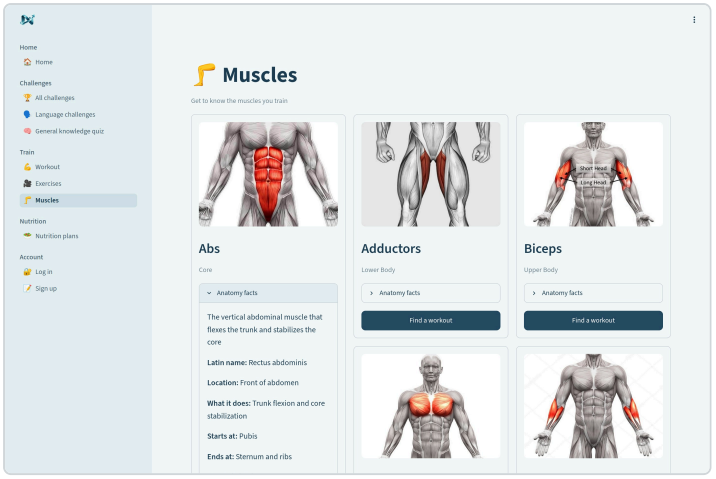
The fitness level calculator (here: Medium), then filter pills for equipment and level, a muscle filter, and a picture from each video.

Exercise video



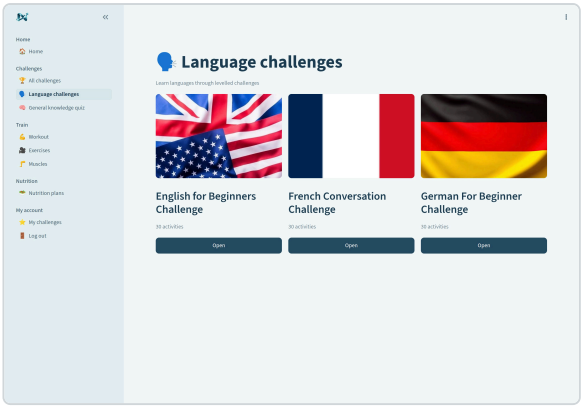
The muscle trained, the equipment, the difficulty, a coaching tip and the video.

Muscles



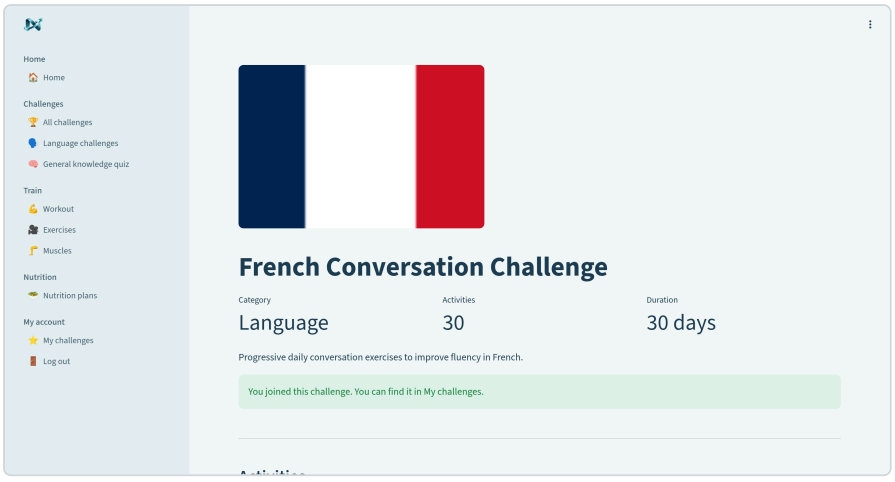
Muscle pictures with an ”Anatomy facts” section that opens on click.

Language challenges



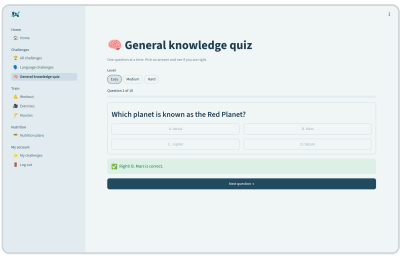
English, French and German with their flags.

Challenge



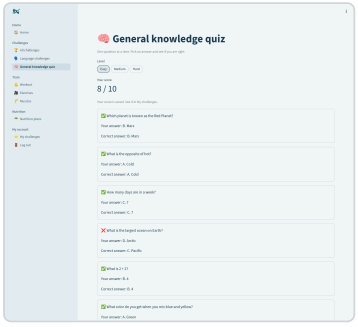
Metrics, the joined message, the progress bar and the activities. A language activity shows the English sentence, the same sentence in the language you learn, and a player that reads it aloud.

Quiz

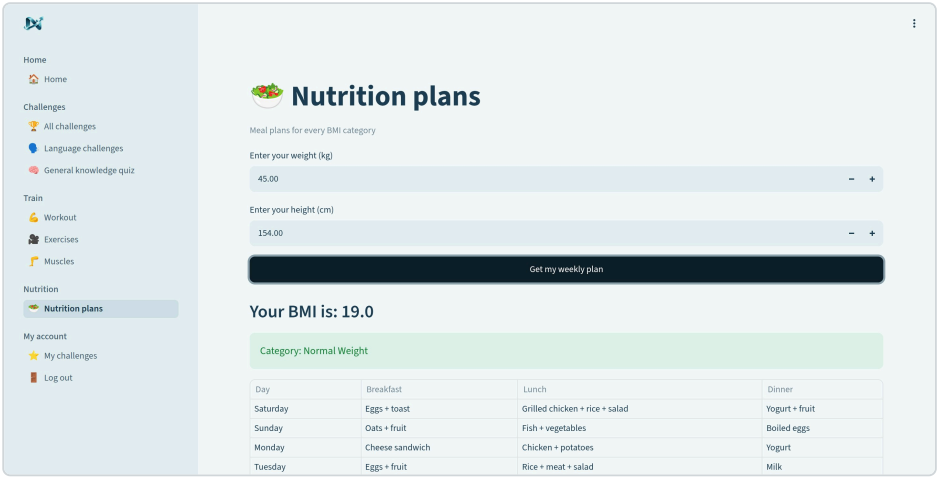


One question at a time, with a progress bar and right or wrong straight away.

Quiz result

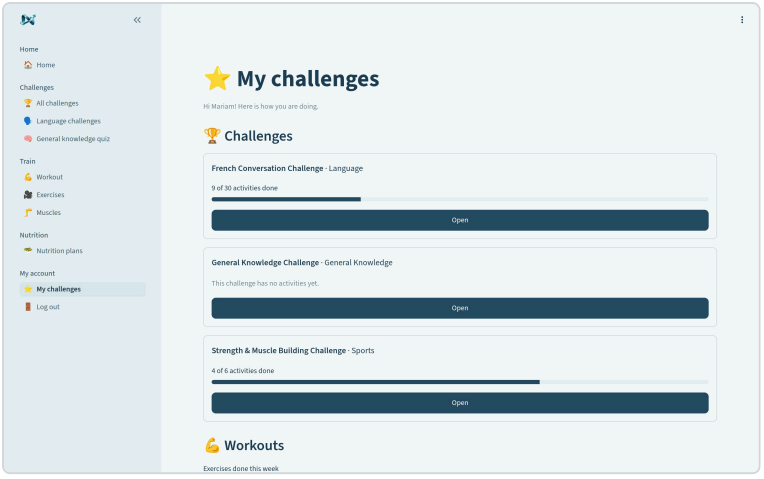


Nutrition plans



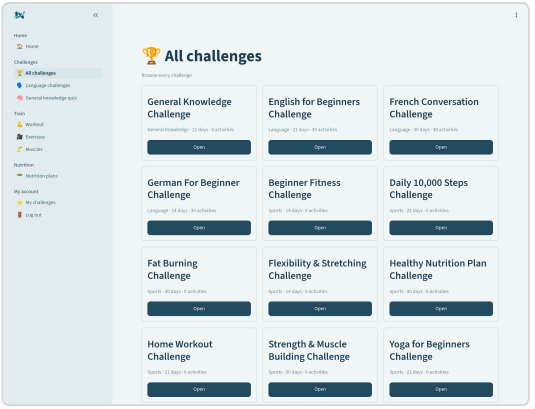
Quiz result	Nutrition plans
The score and a review of every answer.	Your BMI and a 7-day meal plan from your weight and height, then the plan cards with the student's chosen plan marked.

My challenges



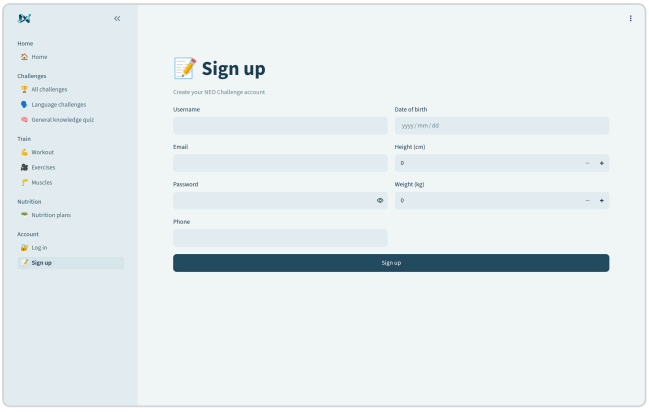
The progress dashboard: challenge progress bars, exercises done this week and quiz scores.

All challenges

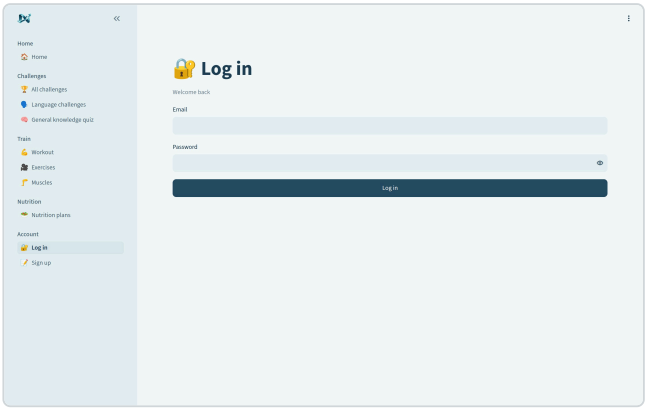


Every challenge with its category, duration and activity count.

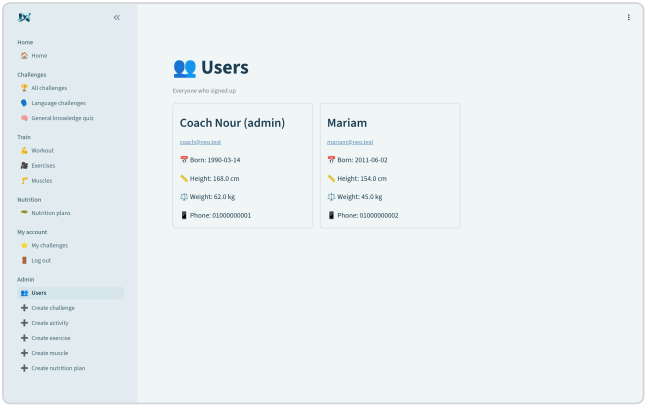
Sign up



Log in

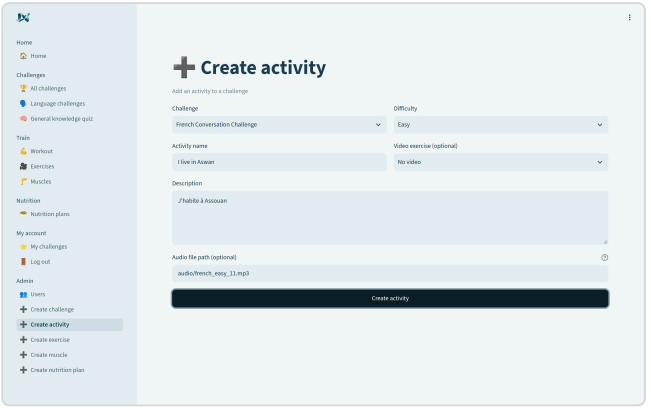


Users (admin)



Everyone who signed up.

Create activity (admin)



Adding a French sentence with its audio file.

Try the app: https://neo-challenge.streamlit.app/

Design System | نظام التصميم

The colours are set once in `.streamlit/config.toml`, so every screen matches without any custom CSS.

Element	Style
Primary Colour	`#254B62`: highlighted buttons and selected options
Background	`#F3F8F9` (main) and `#E4EFF2` (sidebar)
Text	`#1D3E53`
Level Icons	🟢 Easy · 🟡 Medium · 🔴 Hard, so the level is readable without relying on colour alone
Font	Streamlit's default sans-serif in the app. The poster and slides use Bebas Neue (titles), Space Grotesk and JetBrains Mono.
Logo	A teal "N" with an arrow rising out of it, over "NEO CHALLENGE · Better every day" (`assets/logo.png`). The "N" on its own (`assets/logo_mark.png`) is the browser-tab icon and sits at the top of the menu (`st.logo`).
Buttons	Every button is the same filled teal and full width, so the app looks the same everywhere
Cards	Streamlit bordered containers
Layout	Wide layout with a grouped sidebar menu (Home / Challenges / Train / Nutrition / Account or My account / Admin)

Responsive Design | التصميم المتجاوب

Breakpoint	Layout	Tested?
Desktop (1440x900)	Wide layout, sidebar menu, two- and three-column card grids	✅
Tablet (768-1024px)	Same layout with narrower columns	☐
Mobile (< 768px)	Streamlit stacks the columns so each card takes the full width; checked only briefly	⚠️ Partially

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6. Data Sources | مصادر البيانات

Data Sources | مصادر البيانات

Source	Type	Description
`database/database.db`	SQLite Database	Created by the app when it starts; holds everything below
`database/insert_*.sql`	Static (SQL files)	Our starting content: 12 challenges, 108 activities (90 language, 18 sports), 60 trivia questions, 15 muscles, 60 exercises, 15 nutrition plans
Sign up form	User Input	Username, email, password, phone, date of birth, height, weight
Start challenge / Choose this plan	User Input	Saved in `user_challenges` and `user_nutrition_plans`
Admin forms	User Input	New challenges, activities, exercises, muscles and nutrition plans
Progress	User Input	Ticked activities (`user_activities`), exercises done (`exercise_logs`) and finished quiz rounds (`quiz_results`)
`assets/`	Static (images)	Category art, flags, muscle pictures, equipment photos
`videos/`	Static (.mp4)	60 exercise demonstration clips, one per exercise

There are **no external APIs**: no accounts to create, no keys to store, and nothing that breaks when another service is down.

Database Schema | قاعدة البيانات



Table	Columns	Description
`users`	id, username, email, password, birth_date, height, weight, phone, ssn, is_admin, created_at	Student accounts; email is unique; the first account has `is_admin = 1`
`challenges`	id, name, category, duration, description, created_at	12 challenges: 8 Sports, 3 Language, 1 General Knowledge
`activities`	id, challenge_id, name, description, difficulty, exercise_id, audio_url, created_at	108 activities: 30 per language challenge (each with an mp3 in `audio_url`) and 18 Sports ones; 12 workout activities link to a video exercise
`questions`	id, challenge_id, question, option_a… option_d, correct_answer, level, created_at	60 trivia questions (10 Easy, 20 Medium, 30 Hard)
`user_challenges`	user_id, challenge_id	Who joined what; one row per user and challenge
`nutrition_plans`	id, bmi, category, goal, calories, breakfast, lunch, dinner, water_intake, created_at	15 plans across 4 BMI categories
`user_nutrition_plans`	user_id, nutrition_plan_id	The plan each user chose; one per user
`user_activities`	user_id, activity_id, done_at	Activities each user ticked Done

Table	Columns	Description
`exercise_logs`	user_id, exercise_id, done_on	Exercises each user did; at most once per exercise per day
`quiz_results`	id, user_id, level, score, total, played_at	Every quiz round a logged-in user finished
`muscles`	id, name, latin_name, description, muscle_group, location, function, origin, insertion, muscle_type, difficulty_to_train, image_url, video_url, created_at, updated_at	15 muscles with anatomy facts and a picture
`exercises`	id, muscle_id, name, description, equipment, difficulty, video_url, thumbnail_url	60 exercises: Machine 18, Free Weights 20, Cable 12, Bodyweight 10

How the database keeps itself correct:

- **Foreign keys are switched on** for every connection, so an activity can only point at a challenge that exists, and deleting a challenge also deletes its activities and questions (`ON DELETE CASCADE`).
- **`CHECK` rules** allow only the right values (for example Easy / Medium / Hard, or the four equipment types), so a typo cannot hide a row from a filter.
- **Counts are worked out, not typed in.** The number of activities in a challenge is counted with SQL every time, so it can never be wrong.
- **The starting content links by name**, for example `(select id from challenges where name = 'General Knowledge Challenge')`, so it works on a brand-new database whatever ids SQLite gives out.

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7. Educational Content | المحتوى التعليمي

Learning Objectives | أهداف التعلم

By the end of this project, the student will be able to:

1. **Build a multi-screen Python web app with Streamlit**: 20 short screens joined by one menu that changes depending on who is logged in.
2. **Design a relational database that protects itself**: 12 SQLite tables with primary keys, foreign keys with cascade deletes, two join tables and `CHECK` rules.
3. **Manage app state and navigation**: use `st.session_state` to remember the logged-in user, the selected challenge or exercise, and the current quiz.
4. **Organise and load real content**: write SQL seed files for 12 challenges, 108 activities, 60 exercises with 60 videos, 60 trivia questions, 15 muscles and 15 nutrition plans that link to each other by name.

5. **Write code other people can read:** keep every file short and plain, explain the "why" in comments, and list every trade-off honestly.

Curriculum Alignment

Skill Area	What the Student Practices
Problem Solving	Turning a real observation ("teenagers use several separate apps for fitness, languages and trivia") into a product decision: keep every task short, free and in one place.
Computational Thinking	Data modelling and SQL: 12 tables, foreign keys, join tables, counting with `group by`, `order by random()` for the quiz, and rules that keep the data correct.
Creativity	The teal colour theme, level icons, and a 60-clip exercise catalogue linked to muscle pictures.
Collaboration	A two-person team working through Git and GitHub: separate commits per feature and merging branches.
Technical Writing	This document, the README with its honest Limits section, and comments that explain the "why" in plain words.

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8. Marketing Plan | خطة تسويقية

Target Audience | الجمهور المستهدف

- **Primary:** students aged 12-18 who want to build small daily habits around fitness, languages and general knowledge
- **Secondary:** teachers, parents and schools who want a free tool to recommend to students
- **Tertiary:** sports clubs and youth / language centres that need material for training sessions

Platforms | المنصات

- Web browser on school computers and laptop labs
- Personal laptops
- Phones (the layout stacks on narrow screens, so the quiz and the exercise videos work on a phone)

Promotion Ideas | أفكار الترويج

1. **Live demo** at the AFFIAT Aswan Finals 2026 and at the school science fair: run the quiz with the audience and show one exercise video.
2. **Short demo video** (about a minute) showing the quiz and one workout video.
3. **Instagram / TikTok posts** with screenshots and a weekly "challenge of the week".

4. **Poster with the tagline** "One app, three daily habits" in school corridors, at the sports club and in the language centre.
5. **Class-vs-class quiz competitions** run by teachers, using the Easy / Medium / Hard levels so every class has a fair chance.
6. **Share with school PE and language teachers**, who can use the exercise library and the language activities in a lesson.

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9. Student Worksheet

My Project Information | معلومات مشروعي

Student Names: Mohamed Abd Al Ghany, Osama Abd Alghany

Team: NEO Team · Techno Future KFS

Project Title: NEO Challenge

What is your project about? | عن ماذا يتحدث مشروعك؟

NEO Challenge is a free web app for students aged 12-18. It puts three daily habits in one place: sports challenges, language activities in English, French and German, and a general knowledge quiz. It also has a workout and exercise library with 60 demonstration videos, a muscle explorer, and nutrition plans matched to BMI category. The tagline is "One app, three daily habits: train your body, grow your languages, sharpen your mind."

What problem does it solve? | ما المشكلة التي يحلها؟

Teenagers split self-improvement across separate adult apps (a fitness tracker, a language app and a quiz app) that are ad-heavy, sometimes paid, and hard to keep a habit with. NEO Challenge is one simple, free place where a student can finish a short challenge, a short language activity or a short quiz every day.

Where does data come from?

From our own SQL files. We wrote `insert_challenges.sql`, `insert_activities.sql`, `insert_general_knowledge.sql`, `insert_muscles.sql`, `insert_exercise.sql` and `insert_nutrition_plan.sql`. When the app starts on an empty database, it creates the tables from `schema.sql` and loads these files by itself. The exercise videos are 60 `.mp4` clips we collected and stored in `videos/`. The 90 language audio clips in `audio/` were made once with gTTS by `tools/make_audio.py`; the app only plays the saved files. There are no external APIs, so the app works with no internet connection.

What data is stored?

12 SQLite tables: 12 challenges (8 Sports, 3 Language, 1 General Knowledge), 108 activities, 60 trivia questions, 15 muscles, 60 exercises each linked to a muscle and a video, 15 nutrition plans, the users who sign up, which challenges they joined and which nutrition plan they chose, and their progress: ticked activities, exercises done each day and every finished quiz round.

What I Learned | ماذا تعلمت

1. **The hardest part of building my project was:** keeping the data correct. Our first database did not switch on foreign keys, so exercises pointed at muscles that no longer existed and activities outlived the challenges we deleted. We rebuilt the database so it checks every link and every value itself.
2. **The most fun part of building my project was:** collecting the 60 exercise videos and linking each one to the right muscle and equipment, and writing the 60 quiz questions.
3. **If I had more time, I would add:** day-in-a-row streaks and reminders, password hashing, and a public link.
4. **One thing I would do differently:** design the database rules before adding content, and keep the code simple from day one. When we rewrote the app so every file reads top to bottom, it became shorter, easier to explain and easier to fix.
